Anuradha Vibhakar

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Education

2017 - 2021 (expected) **DPhil Condensed Matter Physics**
University of Oxford, U.K.

2012 - 2017 **MSci Natural Sciences: Condensed Matter Physics & Inorganic Chemistry**, First (avg. 82%)
Literature review: Magnetic monopoles and spin ice
UCL Global Citizenship Programme: People and the Sea
University College London, U.K.

2009 - 2011 **International Baccalaureate**, 40 points
English (6), Japanese ab. initio. (6), History (6), Physics (7) Maths (6), Chemistry (7)
Hockerill Anglo-European College, U.K

Research Experience

Oct. 17 - present **Magnetism in the triple A-site columnar ordered quadruple perovskites**
Supervisors: Dr. Roger Johnson & Prof. Paolo Radaelli, University of Oxford, U.K.

DPhil: Funded by an EPSRC Studentship. Determining the magnetic phase diagram of a family of newly synthesised quadruple perovskite manganese oxides: using state of the art diffraction based techniques, a symmetry based and group theoretical approach to data analysis and interpretation, and by creating toy magnetic models in Python. Experiments performed: neutron powder diffraction, small-angle neutron scattering, and non-resonant x-ray diffraction. Facilities used: WISH time of flight diffractometer (ISIS), Zoom small angle neutron scattering instrument (ISIS), I16 beamline at (Diamond Light Source).

Sept. 16 - July 17 **Using density functional theory to explore the stability of ferroelectric domain walls**
Supervisor: Prof. David Bowler, London Centre for Nanotechnology & UCL, U.K.

Masters project: Investigated the effect of defects and system size on the stability of 90°domain walls in $BaTiO_3$ using CONQUEST, a linear scaling DFT code using the London Centre for Nanotechnology's high performance computing facilities. The project involved constructing simulation cells and analysing the structure and polarisation of domain walls. I presented my work to academics in the Department of Physics and Astronomy at UCL and submitted a thesis for which I was awarded a grade of 79%.

June - August 16 **Building a database of reactions and collisional processes occurring in plasmas**
Supervisor: Prof. Jonathan Tennyson, Quantemol Ltd, U.K.

Summer work placement: Created database of plasma chemistry to aid plasma process modelling for use in the semiconductor industry. Calculated cross section data for methane, ethane, propane and silane using technical reports. Added to existing code written in Django (Python) to facilitate faster uploads. Used SQL and Python scripts to upload 60% of datasets on database. Attended weekly meetings where I contributed to discussions regarding the progress of the database and recommended possible strategies for improvement. Proof read technical documents to be published by the company.

July - August 14 **Characterising the structural and electrical properties of ferroelectric perovskite oxides**
Supervisor: Dr Pavlo Zubko, London Centre for Nanotechnology & UCL, U.K.

Summer internship: Funded by an ESPRC bursary to characterise the structural and electrical properties of ferroelectric perovskite oxides. Performed thin film X-ray diffraction (XRD). Used Matlab to convert raw XRD data into reciprocal space maps. Conducted atomic and piezo force microscopy (AFM and PFM). Ran capacitance measurements and plotted hysteresis curves. Used Labview to create a program to increment the temperature of a furnace and subsequently run electrical measurements. Built a sample holder to withstand temperatures of above 500°C. Contributed to assembly of magnetron sputtering chamber.

July - August 10 **Calculating the proper motion of L and T dwarfs**
Supervisor: Dr Avril Day-Jones, University of Hertfordshire Astrophysics Dept., U.K.

Nuffield bursary student: Funded by a Nuffield bursary to use astronomical databases to calculate the proper motion of L and T dwarfs, which was compared to the proper motions of white dwarfs within 2000 au to find binary systems. Worked in Linux, using GAIA Starlink, Iraf and Geomap. Found 14 potential binary pairs. Presented at GlaxoSmitheKline and the Big Bang Fair. Awarded a Gold Crest Award by the British Science Association. Finalist of the 2011 British National Science and Engineering Competition.

Additional Relevant Experience

July - August 16 **Objectifs de Dévelopment Durable Summer School**

Prof. Francois Grey, Tsinghua University, China & University of Geneva, Switzerland

Funded by Tsinghua University, University of Geneva, UCL and the Lego Foundation to attend a unique summer school aimed at crowdsourcing sustainable development. Spent a week at the University of Geneva, attending lecturers at international organisations (the WHO, the UN, CERN). Spent two weeks at Tsinghua university, visiting factories and makespaces in Beijing and Shenzhen to learn about manufacturing in China. Learnt CAD and 3D printing. Part of two projects, which we presented at the Confucius Institute Geneva, the Shenzhen Open Innovation Lab and the Cambridge makespace.

openAFM Part of a team building an open-source AFM to be used as an education tool in schools; using a piezobuzzer, a DVD head, an arduino and 3D printed components. Responsible for the mechanics of the project; incorporating a three point kinematic mount onto the stage of the openAFM, so that it can be used with a Bruker multimode AFM.

SHENDY Leading a team of eight to create a low-cost open-source arsenic detector for use in refugee camps and rural villages, where water networks may be contaminated by arsenic. Detector uses an arduino nano connected to a light to frequency sensor. Collaborating with students from Tsinghua university. Secured laboratory space in Chemistry department at UCL to continue working on project.

Conferences

- Talk, American Physical Society March Meeting 2020, Colorado, USA. Recipient of the APS GMAG Travel Award.
- Poster, Quantum Materials Symposium 2019, Oxford, U.K.
- Invited talk and poster, 'Magnetic Crystallography', 53^{rd} International School of Crystallography, Erice, Sicily, 2019.
- Invited talk and poster, Winter Crystallography Meeting 2018, Abingdon, U.K.
- Chaired session and poster, ISIS Student Meeting 2018, Abingdon, U.K.

Publications

Spontaneous rotation of ferrimagnetism driven by antiferromagnetic spin canting.
A. M. Vibhakar, D. D. Khalyavin, P. Manuel, L. Zhang, K. Yamaura, P. G. Radaelli, A. A. Belik, and R. D. Johnson, Phys. Rev. Lett, 124, 127201, (2020). **Editor's suggestion**.

Magnetic structure and spin-flop transition in the A-site columnar ordered quadruple perovskite $TmMn_3O_6$
A. M. Vibhakar, D. D. Khalyavin, P. Manuel, L. Zhang, K. Yamaura, P. G. Radaelli, A. A. Belik, and R. D. Johnson, Phys. Rev. B 99, 104424 (2019).

Helical magnetism in Sr-doped $CaMn_7O_{12}$ *films*
A. Huon, **A. M. Vibhakar**, A. J. Grutter, J. A. Borchers, S. Disseler, Y. Liu, W. Tian, F. Orlandi, P. Manuel, D. D. Khalyavin, Y. Sharma, A. Herklotz, H. N. Lee, M. R. Fitzsimmons, R. D. Johnson and S. J. May, Phys. Rev. B 98, 224419 (2018).

Strain engineering a multiferroic monodomain in thin film $BiFeO_3$
N. W. Price, **A. M. Vibhakar**, R. D. Johnson, J. Schad, W. Saenrang, A. Bombardi, F.P. Chmiel, C. B. Eom and P. G. Radaelli, Phys. Rev. B 98, 224419 (2018).

Mentoring, Outreach and Interests

Co-organiser of the 2019 Conference for Undergraduate Women in Physics, Oxford University, U.K. Ran departmental talks to primary school girls interested in physics (2017-2019). Provided private tutoring and mentoring to three girls studying STEM subjects for GCSEs and A-levels. (2016-2019). Founded and secured funding for a weekly meeting to encourage collaboration between undergraduate women in physics (2019-present). Keen interests in squash, hiking and live music.